

Simulation-informed transfer learning improves low-data nanobody thermal stability prediction

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Experimental measurements of nanobody melting temperature (T_m) are scarce, which limits machine-learning models of thermal stability. Simulations can provide more training labels, but these labels describe properties related to T_m rather than T_m itself. We asked how the choice of simulated variants and measured quantity affects their usefulness. We trained multi-task models that used a shared ESM2 sequence model for experimental T_m and one computed label at a time. We used 57 sequences for training, 114 for validation, and 396 for held-out testing. Each label and encoder setting was tuned separately using only the T_m validation set, then compared on the same test proteins. The choice of simulated variants mattered. An MD native-contact label reduced frozen-encoder T_m error by 0.20°C when computed for local mutation scans on two fixed structures, compared with 0.05°C for a mixed set of sequences. The measured quantity also mattered. Free-energy perturbation (FEP) labels improved both frozen and fine-tuned encoders, reducing the fine-tuned mean absolute error from 6.55°C to 6.40°C. Native-contact labels helped only the frozen encoder, while Rosetta and ThermoMPNN scores gave little or no improvement. These results show that simulation data for experimental prediction should be built from carefully chosen variants and from quantities that are closely related to the experimental property.

nanobody thermostability | melting-temperature prediction | multi-task transfer learning | protein language model | molecular simulation labels

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Significance

Simulations can provide more protein-stability labels than experiments, but those labels do not always improve prediction of an experimental property. For nanobody T_m prediction, two choices mattered: which variants were simulated and which quantity was calculated. Native-contact labels helped more when computed for local mutation scans than for a mixed set of sequences. Free-energy labels helped both frozen and fine-tuned sequence models, whereas native-contact labels helped only the frozen model. More simulation is therefore not enough; the variants and the calculated quantity must be chosen carefully.

Introduction

Machine-learning models for molecular design need measured examples, but many experiments are slow or expensive. One way to increase the training data is to calculate related properties with quantum chemistry, molecular simulation, or other physical models. This approach is often called simulation-to-real (Sim2Real) learning. It has been used in materials science to combine large computed data sets with smaller experimental data sets (1–3). In some cases, prediction improves as more computed data are added (3). This outcome is not guaranteed. A calculation may use an imperfect model, sample the wrong systems, or measure a property that is only weakly related to the experiment.

Protein stability makes this problem clear because several common labels have different physical meanings. Melting temperature (T_m) is the experimental temperature at which folded and unfolded populations are balanced under a given assay. A mutation free-energy change ($\Delta\Delta G$) instead describes the change in folding free energy caused by a sequence change. Rosetta and learned stability models provide scores on their own scales, while molecular dynamics (MD) can provide structural measurements such as the fraction of native contacts that remain during a trajectory. All of these labels concern protein stability, but none is a T_m measurement. Adding more of the wrong label may therefore leave T_m prediction unchanged or make it worse.

We study this question in nanobodies, single-domain VHH antibodies used as small and modular binding proteins (4, 5). Thermal stability affects their expression, storage, and formulation (6–8). Direct measurements and detailed simulations of stability remain costly (9–12). Protein language models such as ESM2 can learn useful sequence features from large protein databases (13–15), and several methods use these features to predict T_m (16–21). However, a sequence model does not remove the need for measured T_m values from the protein family of interest.

Large protein-stability data sets are becoming available. For example, Tsuboyama et al. measured folding stability for about 776,000 natural and designed protein domains by cDNA-display proteolysis (22). Those domains are 40–72 residues long and differ from VHH domains in both sequence and measurement type. Models trained on small domains also transfer less well to

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longer proteins (23). Nanobody-specific data and validation are therefore still needed (24–26).

The open question is not simply whether computed labels can be added to a Tm model. It is which variants should be simulated and which quantity should be calculated. A mixed set of unrelated sequences may contain unwanted differences such as sequence length. A mutation scan on one or two fixed structures gives a more controlled set of sequence changes. Even with the same variants, a mutation free energy, a Rosetta score, and an MD native-contact value may differ in how well they relate to Tm. These choices are made before model training, so they should be tested as part of the simulation plan.

Here, we compare several computed labels in the same low-data Tm task. The labels are mutation free energies from free-energy perturbation (FEP), Rosetta mutation scores, ThermoMPNN scores, Rosetta scores for ESM2-proposed or random variants, and MD native-contact values. For the MD label, we compare a mixed sequence set with local mutation scans on the same 1MEL and 4IDL structures used for FEP. Each computed label is trained as a second task that shares an ESM2 sequence model with Tm prediction (Fig. 1). We tune every label separately using only the experimental Tm validation set. We then compare the selected models on the same held-out Tm test proteins using paired errors. In this paper, “transfer” means that adding a computed label lowers the held-out Tm error.

Two results stand out. First, the choice of simulated variants matters. The MD native-contact label gives a much larger improvement with a frozen encoder when it is calculated for local mutation scans on fixed structures than for a mixed set of sequences. Second, the calculated quantity matters. FEP labels improve Tm prediction with both frozen and fine-tuned encoders, whereas native-contact labels help only the frozen encoder. Rosetta and ThermoMPNN scores give little or no improvement after separate tuning. These results show that useful simulation data require two choices before calculation: a controlled set of sequence changes and a quantity that is closely related to the experimental property.

Materials and Methods

Study design

We asked whether computed protein-stability labels could improve nanobody melting-temperature (Tm) prediction when only a small experimental training set was available. Experimental Tm was always the task used to choose a model. Each computed label was added as a second regression task that shared all or part of the sequence model with the Tm task. We compared each label with a Tm-only model trained and selected in the same encoder setting.

All choices were made with the experimental Tm validation set. The test set was used only after the model

form and training settings had been fixed. Errors on the computed labels were not used for checkpoint or hyperparameter selection.

Experimental Tm data

We used the thermo-seq task from NbBench, a public nanobody benchmark (25). We reassigned its three published partitions before the comparisons in this study so that the smallest partition was used for training. The resulting split contained 57 training, 114 validation, and 396 test sequences. This split models the setting that motivated the study: few Tm measurements for fitting a model, but enough held-out proteins for paired comparisons. The training set fit model parameters, the validation set selected checkpoints and hyperparameters, and the test set was kept for final evaluation.

Tm values were stored in degrees Celsius. A pipeline consisting of a robust scaler followed by min–max scaling to [0,1] was fit on the 57 training values only. It was then applied to the training and validation labels. Predictions were returned to degrees Celsius before validation and test errors were calculated. Validation and test Tm values therefore did not affect the scaling step.

Computed-label tables and sampling

We tested seven sources of computed labels: FEP mutation free energies, matched MD native-contact values, Rosetta mutation scores, ThermoMPNN predictions, Rosetta scores for random variants, Rosetta scores for ESM2-proposed variants, and MD native-contact values for a diverse nanobody panel. These values were treated as related measurements, not as estimates of Tm.

FEP, Rosetta, and ThermoMPNN used two single-mutation tables based on the 1MEL and 4IDL structures. Each contained 435 and 409 rows, respectively. The matched MD tables used the same two structures and mutation scans and contained 431 1MEL rows and 406 4IDL rows that reached the common analysis time. The random and ESM2-proposed variant sets each contained 1000 two-mutation sequences per structure. The diverse MD table contained 1143 single-domain antibody sequences.

For all two-table sources, the notation n means a cap of n rows from each structure table, not n rows in total. Thus, $n = 320$ gives at most 640 computed-label rows. Rows were sampled independently from the 1MEL and 4IDL tables with the run seed. The diverse MD source had one table, so n is the total number sampled from that table. After sampling, each computed-label table was divided at random into 80% training and 20% monitoring rows. These monitoring rows were not passed to the trainer for checkpoint selection; only the experimental Tm validation rows were used for that purpose.

The 1MEL and 4IDL tables were kept as two tasks because their score ranges can differ. Models used either two separate output heads or one output head with a learned structure indicator, as described below. Ev-

ery processed table supplied an amino-acid sequence and one scalar label scaled to $[0,1]$. FEP, Rosetta, and ThermoMPNN scores were sign-reversed so that larger values indicated a more favorable change, then processed separately for each structure with robust scaling, a Yeo–Johnson transform, and min–max scaling. MD contact values were min–max scaled with larger values indicating greater contact retention.

We also checked exact sequence matches between every computed-label table and the NbBench splits. There were no exact matches for the FEP, matched MD, Rosetta, ThermoMPNN, random-variant, or ESM2-proposed tables. The diverse MD table contained 2 training, 4 validation, and 8 test sequences also present in NbBench. We retained these rows and report the overlap because it limits the interpretation of that comparison.

FEP mutation free energies

The FEP labels were mutation free-energy changes, $\Delta\Delta G$, for the 1MEL and 4IDL mutation scans. Positive raw $\Delta\Delta G$ denotes a destabilizing mutation. The learning tables therefore used the sign-reversed value before the scaling described above.

The fixed FEP tables were produced with alchemical simulations in NAMD using the CHARMM36 protein force field and explicit TIP3P water (27–29). The starting coordinates were the 1MEL and 4IDL crystal structures. Mutation systems were prepared with the VMD AlaScan workflow and CHARMM hybrid-residue topologies (30, 31). Representative archived input files show a 2-fs timestep, particle mesh Ewald electrostatics, bonds to hydrogen held fixed, a temperature of 300 K, and a pressure of 1 atm (32, 33).

Each forward and backward transformation covered 20 windows of width $\Delta\lambda = 0.05$ between 0 and 1. A window contained 500,000 steps (1 ns), with its first 100,000 steps assigned to equilibration. The folded-state free energy was combined with a residue-specific unfolded-state reference to obtain $\Delta\Delta G$ relative to the wild type. The present learning pipeline starts from the processed CSV files and does not rerun the FEP calculations.

MD native-contact labels

Both MD labels were based on the smooth native-contact fraction Q (34). For a native contact with reference distance d_{ij}^0 and distance $d_{ij}(t)$ at time t , its contribution was

$$q_{ij}(t) = [1 + \exp\{\beta[d_{ij}(t) - \lambda d_{ij}^0]\}]^{-1},$$

with $\beta = 50 \text{ nm}^{-1}$ and $\lambda = 1.8$. Native contacts were pairs within 0.45 nm in the reference frame and separated by more than three residues. Q was the mean over the selected contacts and frames.

Matched mutation scans. We simulated the 1MEL and 4IDL wild types and the same single-mutant sequences used in the FEP comparison. Each protein was

placed in a cubic box of OPC water with a 15-Å buffer and 0.15 M salt, and was described by Amber ff19SB with hydrogen-mass repartitioning (35, 36). OpenMM was used for minimization, equilibration, and production (37). The recorded run settings were 0.2 ns NVT followed by 0.8 ns NPT at 300 K, 0.5 ns restrained NVT at 400 K, and restraint-free NVT production at 400 K. All stages used a 4-fs timestep; production coordinates were written every 10 ps. Although some runs continued to 100 ns, the label used the first 4000 production frames (40 ns) for every variant.

For this source, the reference distances came from production frame 0 and all backbone atoms selected by MDAnalysis were eligible for contacts. We computed $\Delta Q = Q_{\text{mutant}} - Q_{\text{WT}}$ within each structure and then min–max scaled it. Subtracting one constant wild-type value does not change a min–max-scaled label, so scaled Q and scaled ΔQ are identical within each structure table.

Diverse nanobody panel. The comparison panel was built from SAbDab structures (38). One verified heavy-chain nanobody was taken per PDB entry when an IMGT-renumbered structure was available. Preparation used PDBFixer and pH-aware protonation with pdb2pqr/PROPKA when available (39, 40). The systems used ff19SB, OPC water, 0.15 M salt, and OpenMM. After 300 K NVT and NPT equilibration, two restrained 400 K NVT stages preceded 100 ns of unrestrained 400 K NVT production. Requiring a usable topology and trajectory, a sequence of at least 50 residues, and at least one selected contact gave 1143 rows.

The diverse-panel label used production frame 0 as its reference and averaged approximately the last 30 ns. Its contact set contained backbone heavy-atom pairs for which at least one residue was Asp, Glu, Gln, Asn, Arg, Lys, or His. It was min–max scaled across the single table. The matched and diverse studies used similar simulation time per sequence and therefore compare two ways of spending a similar computational budget. They used the same Best–Hummer form and the same temperature, although their contact selection and averaging window were not identical. We therefore treat this as a comparison of complete simulation plans, rather than as a one-variable test of sequence choice alone.

Rosetta, ThermoMPNN, and variant-selection comparisons

Rosetta scores were calculated with the ddg_monomer application and the ref2015 score function (41, 42). The archived command used 50 iterations, Cartesian minimization, minimum-score aggregation, and full-atom input. The same settings were used for the 1MEL/4IDL mutation tables and for the two sets of two-mutation variants.

For the random set, mutation positions and replacement amino acids were sampled uniformly, exclud-

ing cysteine as a replacement, until 1000 unique two-mutation variants had been obtained for each structure. For the ESM2-guided set, the 650M-parameter ESM2 model generated 100,000 two-mutation variants per structure (14). Variants were ranked by pseudo-log-likelihood, and the top 1% were scored with Rosetta. ThermoMPNN predictions were made for the same 435 and 409 single-mutation sequences used by the structure-matched tables (43). The current loader reads the two processed ThermoMPNN tables, rather than the larger raw Colab outputs stored with them.

Sequence model and regression heads

The main encoder was the 8M-parameter ESM2 model facebook/esm2_t6_8M_UR50D (14). Sequences were padded or truncated to 160 residues. The first-token representation was used as the sequence vector. In the shared model, this vector passed through linear layers of size 256, 128, and 32, with ReLU activation and dropout after the first two layers. The Tm head and each computed-label head were separate 32-to-1 linear layers.

The model search also included residual and latent forms. Both gave Tm and computed-label examples separate 256–128–32 paths. The residual form added a sigmoid-gated correction to a base Tm prediction. The latent form predicted a Tm correction from the Tm vector, the second path’s vector, and their elementwise product. For two-table sources, we compared separate 1MEL and 4IDL heads with a single head conditioned on a learned structure embedding. In the residual and latent implementations, the computed-label loss received a detached encoder vector by default, whereas the Tm path could still update the encoder. In the shared form, losses from all tasks could update the encoder when it was not frozen.

We tested two encoder settings. In the frozen setting, all ESM2 parameters were fixed and sequence vectors were computed once before head training. In the hot setting, all ESM2 parameters could change. AdamW used a smaller learning rate for ESM2 than for the newly initialized layers. Selected controls used 35M- and 650M-parameter ESM2 encoders.

Loss and model selection

Each row had one task identifier and contributed loss only through that task’s output head. We used Huber loss with $\delta = 1$ on the scaled labels. By default, the losses were combined with learned uncertainty weights (44). For task k , loss L_k , and learned log-scale s_k , the term added to the total loss was

$$\frac{L_k}{2\exp(2s_k)} + s_k.$$

Fixed task weights were tested as controls.

Models were trained with AdamW, cosine learning-rate decay, a batch size of 16, 100 warmup steps, and at

most 400 epochs. Mixed-precision training was used on CUDA devices. A checkpoint was saved and evaluated once per epoch. The checkpoint with the lowest mean squared error on the 114 experimental Tm validation rows was retained, and training stopped after 15 epochs without improvement.

Every label source was tuned separately in the frozen and hot settings. The search had two stages. Stage 1 crossed three model forms (shared, residual, and latent) with two head forms (separate and structure-conditioned); Tm-only models searched the three model forms without a computed-label head. Stage 2 held the chosen form fixed. For hot models it tested encoder learning rates of 1×10^{-4} , 5×10^{-5} , 3×10^{-5} , and 1×10^{-5} . For frozen models it tested head learning rates of 1×10^{-3} , 5×10^{-4} , and 2×10^{-4} . Both searches crossed dropout values of 0.05, 0.15, and 0.30 with weight decay values of 0.02 and 0.10.

Each candidate was ranked by the MAE of a three-seed ensemble on the experimental Tm validation set, using $n = 320$ rows per structure table. The selected setting was then trained with five seeds for the final test result. Label-count curves used the same selected setting at $n = 20, 80, 160$, and 320 per structure; they were not retuned at each point.

Test evaluation and statistics

For each condition, the five models predicted all 396 test sequences. Their predictions were averaged in scaled space and then returned to degrees Celsius. The main measure was mean absolute error (MAE). We also calculated root mean squared error, R^2 , and Spearman and Pearson correlations.

Uncertainty was estimated by nonparametric bootstrap over test proteins. For one condition, we resampled its 396 absolute errors with replacement 10,000 times, using random seed 42, and calculated MAE for each sample. The reported 90% confidence interval is the 5th to 95th percentile range.

For a comparison, the same resampled test indices were applied to the candidate and reference error vectors. We calculated ΔMAE as candidate MAE minus reference MAE for every sample. Negative values therefore favor the candidate. Main figures report the mean paired ΔMAE and its 90% confidence interval. We do not use a separate p-value threshold for the main claims.

Software and saved outputs

The data loader, training code, model settings, and ensemble evaluation are in prepare.py, train.py, and the scripts distributed with the repository. Each final run stores its command-line settings, predictions, absolute errors, summary measures, and bootstrap results in machine-readable files. The figure scripts read those saved files directly. The released workflow begins with the fixed processed CSV tables; it does not repeat the raw FEP, Rosetta, ThermoMPNN, or MD calculations.

Results and Discussion

Computed labels were judged only by experimental Tm. We first set up a common test for all computed labels (Fig. 1). Experimental Tm was the prediction target. FEP, MD, Rosetta, and ThermoMPNN values were separate regression tasks; they were not treated as extra Tm measurements. Each task used the same ESM2 sequence model and either shared or task-specific regression layers. We tested a frozen ESM2 encoder and a fully fine-tuned, or hot, encoder.

Only experimental Tm was used to choose a model. The 57 training proteins fit the model, the 114 validation proteins selected the architecture and training settings, and the 396 test proteins were used after selection. Every computed label was tuned separately in the frozen and hot settings. We then compared absolute errors on the same test proteins. This was necessary because a model can predict a computed value well without improving Tm.

The selected Tm-only model had a test MAE of 7.23°C with a frozen encoder and 6.55°C with a hot encoder. The lower error of the hot baseline shows that fine-tuning ESM2 was useful even with only 57 measured training proteins. Against these two baselines, we asked two questions. First, does the set of sequences chosen for simulation affect transfer? Second, does the computed quantity affect whether the benefit remains when ESM2 is fine-tuned?

Local mutation scans gave a stronger MD result

Figure 2a compares two choices made before running MD. The heterogeneous screen used unrelated single-domain antibody sequences of different lengths. The matched scan used local mutations of the fixed 1MEL and 4IDL structures. Both calculations used 400 K MD and the Best-Hummer native-contact fraction Q . They also used similar simulation time per sequence. Thus, the comparison asks how a similar amount of simulation should be distributed over sequence space.

The heterogeneous and matched studies were run and tuned as separate series, so their absolute MAEs cannot be compared directly. Figure 2b instead shows the paired change from the Tm-only model in each series. In the heterogeneous series, the MD label changed MAE by -0.049°C with a frozen encoder (90% CI, -0.085 to -0.012°C) and by $+0.117^\circ\text{C}$ with a hot encoder (90% CI, $+0.012$ to $+0.224^\circ\text{C}$). The frozen gain was small, and the hot model became worse. In the matched series, the frozen change was -0.195°C (90% CI, -0.336 to -0.055°C), about four times the change in the heterogeneous series. The hot change was $+0.029^\circ\text{C}$, with a confidence interval that included zero.

Several differences between the two sequence sets may contribute to this result. The heterogeneous set ranged from 58 to 461 residues, and its Q value had a Pearson correlation of $r = +0.13$ with sequence length. Length could therefore provide an unwanted shortcut. The

matched scans kept length fixed within each structure and supplied local directions: one mutation and its change in contact retention. This is a simpler learning problem. However, the present data do not show that length alone caused the difference. The two MD pipelines also differed slightly in contact selection and in the trajectory frames used for averaging, as detailed in Materials and Methods. The result therefore compares the two complete simulation plans, not a single changed variable.

Wild-type referencing does not explain the matched result. Its label was $\Delta Q = Q_{\text{mutant}} - Q_{\text{WT}}$, but the same wild-type constant was subtracted from every mutation in a structure table. The following min-max scaling removes this constant, so scaled Q and scaled ΔQ are identical within that table.

The frozen label-count curves give a second view of the matched scan (Fig. 2c). Here n is the number of labels sampled from each structure table. FEP MAE changed from 7.13°C at $n = 20$ to 7.01°C at $n = 320$. Matched MD changed from 7.32 to 7.03°C and fell below the 7.23°C Tm-only baseline at the largest setting. The four points are not monotonic enough to support a scaling law, but both sources were better than Tm-only at $n = 320$. Their paired changes were -0.22°C for FEP and -0.20°C for MD. The direct FEP-minus-MD difference was only -0.03°C (90% CI, -0.21 to $+0.16^\circ\text{C}$). Thus, with a frozen encoder, the matched MD scan performed as well as FEP at the largest tested label count.

The main conclusion from Fig. 2 is narrow. Choosing local mutations on fixed structures made the frozen MD result much stronger and removed the hot penalty seen for the heterogeneous screen. It did not make the MD label helpful in the hot model. Sequence choice matters, but it is not the only choice that matters.

Only FEP helped both frozen and hot models

We next compared all computed labels after separate tuning (Fig. 3a). With a frozen encoder, FEP and matched MD gave the clearest gains. FEP reached 7.01°C MAE, a paired change of -0.221°C from Tm-only (90% CI, -0.366 to -0.077°C). Matched MD reached 7.03°C, a change of -0.195°C . ThermoMPNN was third at 7.09°C, but its change of -0.141°C had a confidence interval that included zero. Plain Rosetta, random variants scored by Rosetta, and ESM2-proposed variants scored by Rosetta were close to or worse than the Tm-only model.

The order changed with a hot encoder. FEP reached 6.40°C, the lowest MAE in the study. Its paired change from the 6.55°C Tm-only model was -0.153°C (90% CI, -0.295 to -0.009°C). No other computed label improved the hot baseline. Matched MD, ThermoMPNN, plain Rosetta, and random variants plus Rosetta had changes from $+0.029$ to $+0.144^\circ\text{C}$, all with intervals that included zero. ESM2-proposed variants plus Rosetta were worse by $+0.411^\circ\text{C}$ (90% CI, $+0.181$ to $+0.651^\circ\text{C}$).

The focused comparisons in Fig. 3b separate the two all-atom labels. FEP and matched MD were not distinguishable with a frozen encoder, as noted above. With a hot encoder, FEP was better than MD by a paired -0.182°C (90% CI, -0.359 to -0.009°C). This difference also appears across label counts (Fig. 3c). At $n = 20$, matched MD was worse than hot Tm-only by $+0.47^{\circ}\text{C}$ (90% CI, $+0.29$ to $+0.65^{\circ}\text{C}$). Its error then fell from 7.02 to 6.58°C as n increased, but at $n = 320$ it had only returned to the Tm-only baseline. FEP fell from 6.57 to 6.40°C and was better than baseline at $n = 320$. These four-point curves show recovery and crossover; they do not establish a power law.

FEP $\Delta\Delta G$ and Tm are different measurements. $\Delta\Delta G$ gives the change in folding free energy caused by a mutation, whereas Tm is the temperature at which folded and unfolded states are balanced under an experimental condition. They are nevertheless linked by protein folding thermodynamics. In this data set, the measured Tm examples supply absolute temperature values across nanobodies, while the FEP scans supply relative information about which local sequence changes stabilize or destabilize two structures. The matched MD scan also supplies local information, but contact retention is a more indirect measure of folding stability. This difference is a plausible reason why the two labels tie in the frozen model but separate in the hot model.

We describe this pattern as a difference in transfer depth. Here, “deeper” means only that a benefit remains when the pretrained encoder is allowed to change. A recent preprint from our group gives a more concrete basis for this interpretation (45). That study used sparse autoencoders to examine ESM2 after fine-tuning on nanobody Tm. It found interpretable features linked to surface charge, hydrophobic-core packing, the VHH tetrad, and disulfide bonds. Thus, fine-tuning on nanobody Tm can produce a representation with specific, physically meaningful stability features rather than only a better regression fit.

Together with that result, the present frozen-hot difference suggests a testable hypothesis: mutation free energies may support the same kinds of local stability features that a Tm-fine-tuned encoder learns, while a single contact-retention value may provide a weaker guide for updating them. This remains an interpretation, not a result measured here. We did not apply sparse autoencoders or compare hidden states in the present models, and the selected FEP and MD models did not always use the same regression architecture. A direct next test is therefore to compare the sparse features learned by Tm-only, Tm+FEP, and Tm+MD hot encoders and ask which stability features appear, disappear, or change strength.

Controls, scope, and next tests

Several controls limit alternative explanations. Increasing ESM2 from 8M to 35M or 650M parameters did not

reproduce the 8M FEP result (Supplementary Fig. 4), so a larger pretrained model was not a substitute for the computed label. An analytical correction for periodic-image effects in charge-changing FEP mutations had almost no effect on the scaled labels. Tests that clipped the largest FEP values or added the available Phe scan did not improve transfer. These checks support the use of the original FEP tables, although they do not prove that every large $\Delta\Delta G$ value is quantitatively accurate.

The other label sources provide an important negative result. ThermoMPNN gave at most a weak frozen gain, and none of the Rosetta sources improved both encoder settings. ESM2-proposed variants scored by Rosetta did not beat random variants scored by the same method. We therefore treat these sets as comparisons of sequence choice and score type. The results do not support a claim that the ESM2 proposals form a successful protein-design loop.

The findings agree with the main idea of Sim2Real learning: computed data are useful when they share information with the measured task (3). The protein case adds an earlier decision. Before a label can transfer, the simulated sequences must make the desired relationship easy to learn. A large mixed set is not automatically better than a smaller local scan. After the sequences are chosen, the computed quantity still matters. FEP and matched MD used roughly similar simulation time per variant here, but we did not record wall-clock cost in a form that supports a general cost claim.

This study has several limits. The Tm training set contains only 57 nanobodies, and the matched labels cover only two structures. The heterogeneous and matched MD pipelines differ in more than their sequence sets, and the heterogeneous table includes eight exact matches to the test set. The source list does not cover every useful simulation measurement. The confidence intervals also show that small differences among the middle-ranked labels remain uncertain. Finally, we did not make new Tm measurements or test predictions prospectively.

The next test should extend matched mutation scans to more nanobody structures while keeping the simulation and label-extraction settings fixed. It should compare several physical quantities on exactly the same variants and then test the selected Tm model on new experimental measurements. Such a study would separate sequence choice from label choice more cleanly and show whether the FEP result extends beyond 1MEL and 4IDL.

Conclusion

Computed labels improved low-data nanobody Tm prediction only in specific settings. The sequences chosen before simulation mattered: MD native-contact labels from local mutation scans on 1MEL and 4IDL gave a clearer frozen-encoder gain than labels from a heterogeneous sequence screen. The calculated quan-

tity also mattered. FEP mutation free energies improved both frozen and fine-tuned models, whereas matched MD native-contact values improved only the frozen model; Rosetta and ThermoMPNN scores gave no clear gain. Simulation data for experimental prediction should therefore be planned around controlled sequence changes and a quantity closely related to the measured property, rather than around data volume alone. The next test is to extend matched scans to more structures and evaluate the resulting models on new T_m measurements.

In the longer term, the same idea could guide an iterative design study. A model could choose which variant to calculate next, receive an FEP or MD result, and use that result to choose the following variant. Reinforcement learning is one way to make these sequential choices, but a computed score alone would not be a reliable reward. Our ESM2-proposed variants did not outperform random variants when both were scored by Rosetta, showing that a generator and a physics-based score do not by themselves make a useful design method. Any future reinforcement-learning study should therefore be compared with random sampling and simpler active-learning methods, and its success should be judged prospectively by new experimental T_m measurements.

Conflict of interest

T.M. is an employee of Epsilon Molecular Engineering, Inc. Y.M. is a scientific advisor of Epsilon Molecular Engineering, Inc.

Author contributions

Y.M. conceived and designed the study. Y.M. and T.M. carried out the overall computational study and analyzed the data. S.O. and K.O. performed the FEP calculations. Y.M. and T.M. wrote the manuscript. All authors discussed the results, reviewed the manuscript, and approved the final version.

Data availability

The source code for data processing, model training, analysis, and figure generation is available at <https://github.com/ymatsunaga/sim2real-nanobody-tm>. The processed data used in this study, together with reduced MD trajectories sufficient to reproduce the reported labels, will be available from Zenodo (DOI: 10.5281/zenodo.XXXXXXX). Full raw simulation files are not included because of their size and are available from the corresponding author upon reasonable request.

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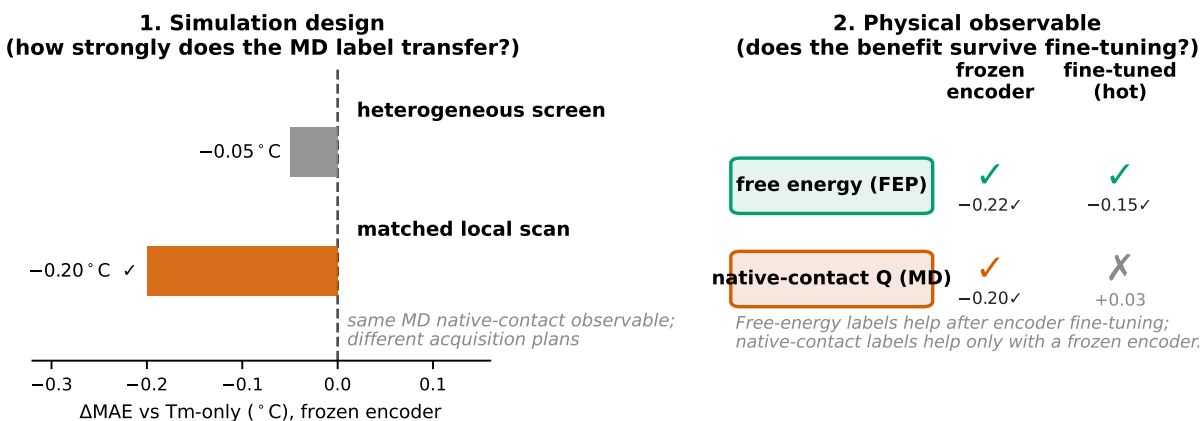
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Two choices affect how simulation labels improve low-data nanobody melting-temperature (T_m) prediction



Graphical abstract. Two choices affect whether simulation labels improve low-data nanobody T_m prediction: which variants are simulated and which quantity is calculated. Matched local mutation scans make MD native-contact labels more useful. Free-energy labels help both frozen and fine-tuned encoders, whereas native-contact labels help only a frozen encoder.

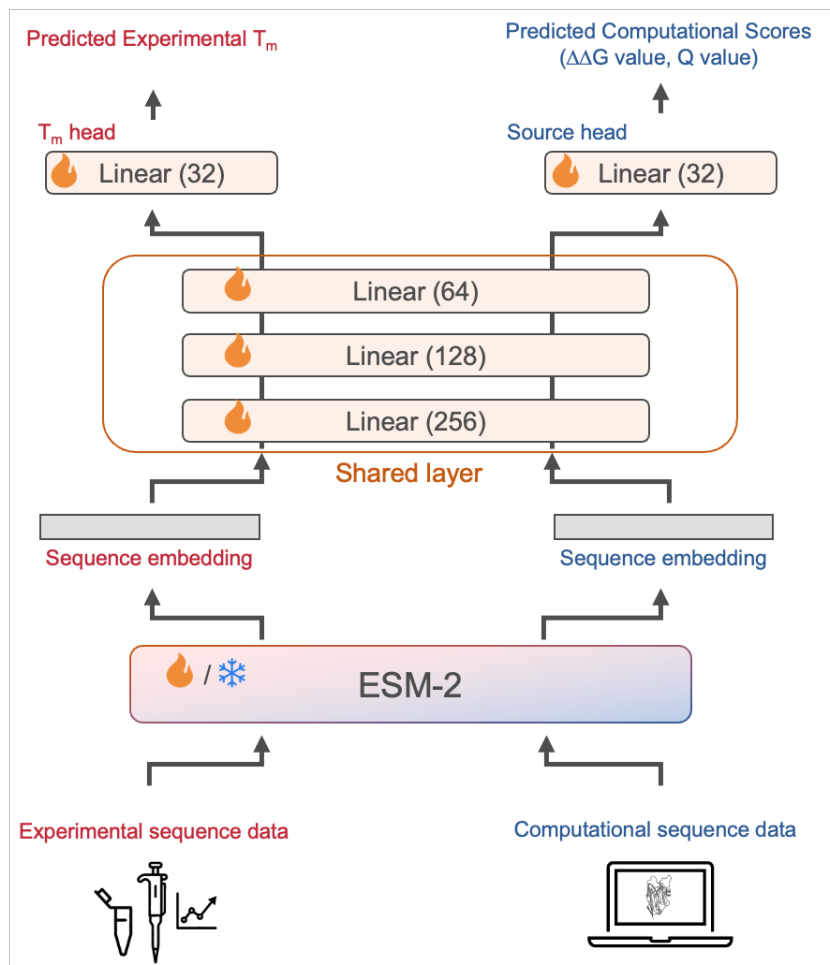


Fig. 1. Multi-task model used to test computed stability labels. Experimental sequences and sequences with computed labels pass through the same ESM2 model and shared multilayer perceptron. Separate scalar heads predict T_m and the computed label. The ESM2 model is either kept frozen or fine-tuned. Models are selected using experimental T_m validation data, and the final comparison uses held-out T_m test data.

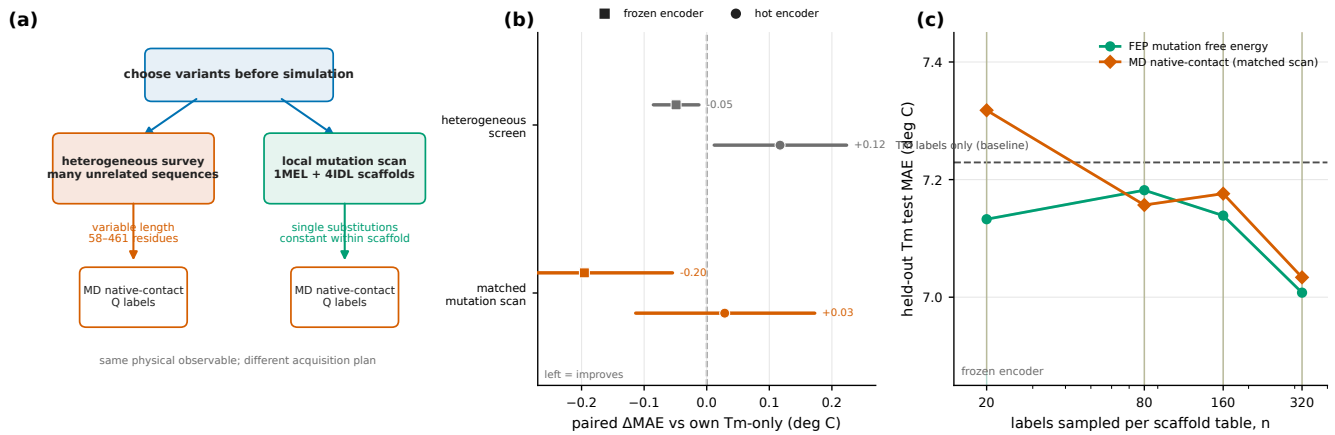


Fig. 2. Simulation design changes the value of an MD native-contact label. (a) A heterogeneous screen of variable-length sequences and local mutation scans on the fixed 1MEL and 4IDL structures use similar simulation time per sequence and produce labels from the Best-Hummer Q family. (b) Paired Δ MAE for each design and encoder setting. Because the two series were tuned independently, each value is relative to that series' own Tm-only model; their absolute MAEs are not compared. The matched scan gives a larger frozen benefit and avoids the hot penalty, but gives no hot benefit. (c) Frozen-encoder MAE as the number of FEP or matched-MD labels increases. The dashed line is the selected frozen Tm-only model. Here n is the number sampled from each structure table. Points and bars show ensemble MAE and 90% bootstrap confidence intervals.

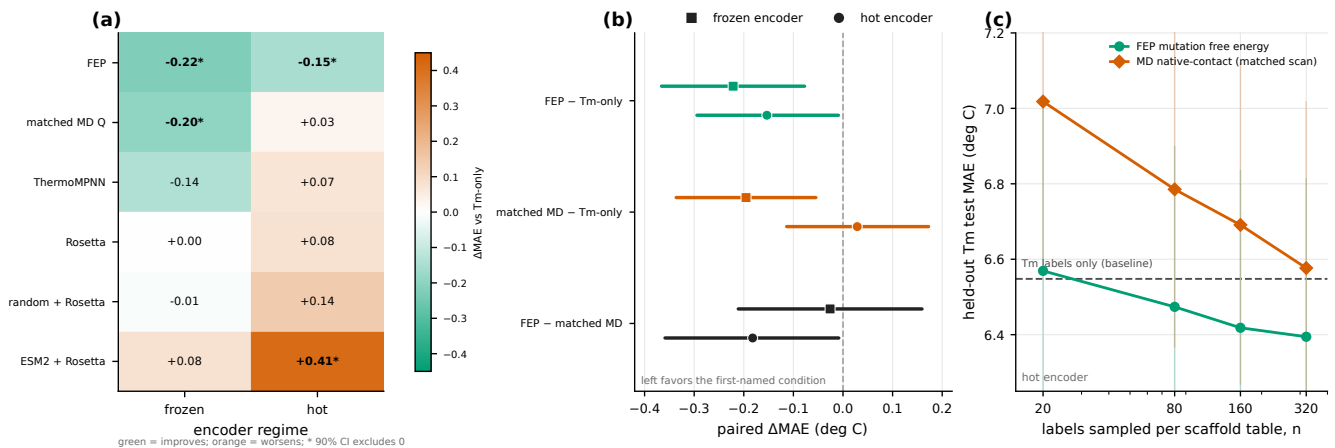


Fig. 3. FEP is the only computed label that improves both encoder settings. (a) Paired Δ MAE from the selected Tm-only model for every tuned label. Green denotes lower error and orange higher error. An asterisk marks a 90% paired bootstrap confidence interval that excludes zero. (b) Paired comparisons of FEP and matched MD with Tm-only, and direct FEP-minus-MD comparisons, in the frozen and hot settings. Values left of zero favor the first-named condition. (c) Hot-encoder MAE as FEP or matched-MD labels increase. FEP becomes better than Tm-only, whereas matched MD recovers from its low-count loss only to the baseline. Here n is the number sampled from each structure table. Bars are 90% bootstrap confidence intervals over the 396 test proteins.

Supplementary Methods

Experimental target data and target-label scaling

All target-task comparisons use the same NbBench thermo-seq endpoint (25). The processed tables hold an amino-acid sequence field and a melting-temperature label in degrees Celsius. The split sizes are 57 training, 114 validation, and 396 held-out test sequences. As described in Methods, we deliberately assigned the smallest NbBench partition to training and the largest to the test set so that the model sits in the low-data regime the study targets.

Split definitions. The splits are fixed throughout. The 57 training examples fit network parameters, the validation split drives checkpoint selection, early stopping, and choice among candidate settings, and the test split stays untouched until those choices are fixed. The same rule applies to Tm-only models and to models trained with computational labels.

Target-label scaling. Before optimization, the training workflow maps the experimental Tm values to a scaled regression target. The scaler is fit only on the 57 training labels, using robust scaling followed by linear rescaling to [0,1]. Validation labels are transformed with this training-fitted scaler during model selection. Test predictions are averaged across seeds in scaled-label space and inverse-transformed to degrees Celsius before errors are computed, so the validation and test label distributions never influence the fitted transform.

Experimental-label scaling reference. This reference uses the same training split but subsamples it to the requested number of labels with the run seed; the validation and test splits are never subsampled. It serves only as a positive control for the pipeline. Adding true target labels should lower the Tm prediction error.

Processed computational-label tables

Computational labels are loaded as computational tasks rather than appended to the Tm data as extra measurements. Table 1 lists the processed target and computational-label files used in the reported comparisons. Mutation-effect tables carry sequence fields and [0,1]-scaled computational-label fields. The MD native-contact tables use the same processed format so that they can enter the same computational-task loader; there, the scaled label is a [0,1]-scaled native-contact fraction rather than a mutation free energy. The FEP-matched MD table (the primary MD label) is built on the same 1MEL/4IDL mutation scans as FEP; the diverse-panel MD table provides the heterogeneous simulation-acquisition comparison.

Table 1. Processed target and computational-label data. Row counts exclude header rows.

Data type	Role	Processed rows	Use in reported comparisons
Experimental Tm	Target train	57	Model fitting
Experimental Tm	Target validation	114	Checkpoint and setting selection
Experimental Tm	Target test	396	Final held-out evaluation
FEP mutation free energy	Computational tasks	435 + 409	Main computational task
MD native-contact (FEP-matched scan)	Computational tasks	431 + 406	Matched-design MD label (primary)
Rosetta mutation score	Computational tasks	435 + 409	Computational-label comparison
ThermoMPNN stability score	Computational tasks	435 + 409	Computational-label comparison
Random variants scored by Rosetta	Computational tasks	1000 + 1000	Computational-label comparison
ESM2-proposed variants scored by Rosetta	Computational tasks	1000 + 1000	Computational-label comparison
MD native-contact (diverse panel)	Computational task	1143	Heterogeneous simulation-design comparison

Computational-task assignment. The loader assigns each row to a target or computational task. Experimental Tm rows use the Tm head. In the main mutation-effect model, the two template-specific tables use separate computational heads, and the MD-derived Q-value table uses a single computational head. This assignment fixes which regression head an example uses and which task-specific loss term it contributes to.

Sequence-overlap check. We checked exact sequence overlap between the processed computational tables and the fixed NbBench splits. The FEP, Rosetta, and ThermoMPNN mutation-effect tables, the FEP-matched MD native-contact table, and the random and ESM2-proposed variant tables — all built on the 1MEL and 4IDL scaffolds — had no exact match to any NbBench training, validation, or test sequence. The diverse-panel SAbDab MD table contained exact matches to 2 NbBench training, 4 validation, and 8 test sequences. We disclose this because it bears on the simulation-design comparison. The diverse panel gives a small frozen-encoder gain, so the overlap is a limitation of that secondary effect and is one reason not to interpret the matched-versus-heterogeneous contrast as a fully isolated causal experiment. It cannot explain the matched-scan result, whose sequences have no exact NbBench overlap.

Computational-label sampling. For mutation-effect experiments with requested count n , the loader samples up to n rows independently from each of the two template tables using the run seed; the final FEP setting with $n = 320$ therefore supplies 320 sampled 1MEL rows and 320 sampled 4IDL rows before the computational-task split. The FEP-matched MD native-contact label samples from the same two template-specific tables. The diverse-panel MD control samples n rows from its single table. Computational-label rows are split 80/20 into computational-training and computational-validation rows with the same seed. The production training call then filters the validation data passed to the trainer so that checkpoint selection and early stopping depend only on the experimental Tm validation examples.

Diverse-panel MD control: SAbDab structure selection

The following describes the diverse-panel MD native-contact comparison used for the simulation-design contrast in Fig. 2. The FEP-matched MD label used for the main result is built on the 1MEL/4IDL mutation scans as described in Methods. The diverse-panel MD control was built from SAbDab, the Structural Antibody Database (38). The archived SAbDab nanobody summary table holds 2422 rows with PDB identifier, heavy- and light-chain assignments, model index, antigen annotation, experimental method, resolution, species, and other fields. The manifest workflow matches this summary against the archived IMGT-renumbered PDB files.

Structure-selection rule. The rule is deterministic. For each PDB identifier with both a SAbDab summary row and an IMGT-renumbered PDB file, the workflow takes the first heavy-chain row in the SAbDab table and accepts the chain only if that chain identifier is present in the IMGT PDB file. The output manifest holds one entry per PDB identifier, with fields for the manifest row number, PDB identifier, selected heavy chain, SAbDab TSV row, IMGT PDB path, and MD run directory. This initial one-per-PDB manifest contained 1177 structures.

400 K simulation panel. The 400 K simulations reused the systems prepared for the 300 K runs and built a 1146-entry 400 K manifest, consisting of 798 X-ray, 345 electron-microscopy, and 3 NMR structures according to the SAbDab method field. None of the selected rows had an annotated light chain, consistent with a single-domain antibody panel. After trajectory and Q-value processing, 1143 unique PDB identifiers yielded usable computational-label rows.

MD system preparation and simulation protocol

System preparation. Each structure was prepared as a nanobody-only system. The workflow takes the IMGT-renumbered PDB as the input structure, extracts the selected heavy chain, keeps only protein atoms, and leaves the termini uncapped, at pH 7.4. Chains are cleaned with PDBFixer, and pH-aware protonation is assigned through pdb2pqr/PROPKA when available (39, 40), with a fallback to AmberTools-based hydrogen assignment and Amber naming. The prepared chain is solvated in explicit OPC water with 15 Å padding and 0.15 M NaCl, and Amber topology and coordinate files are built with ff19SB and OPC (35, 36, 46).

300 K equilibration branch. The base 300 K branch uses OpenMM on CUDA devices (37). It runs 1 ns of NVT followed by 1 ns of NPT at 300 K and 1 atm, with a 2-fs timestep and no hydrogen-mass repartitioning. Production at 300 K was generated for the broader study, but the computational label used here comes from the 400 K branch.

400 K production branch. The first 400 K relaxation stage restarts from the 300 K equilibrated state and runs 1 ns of NVT at 400 K with $C\alpha$ restraints; the second repeats 1 ns of restrained 400 K NVT. Both relaxation stages use a 2-fs timestep and no hydrogen-mass repartitioning. Production runs 100 ns of restraint-free 400 K NVT, with hydrogen-mass repartitioning, 4 amu hydrogen mass, a 4-fs timestep, and coordinate/energy output every 100 ps.

Integrator and reporting settings. In OpenMM, explicit-solvent periodic systems use PME electrostatics with a 1.0-nm nonbonded cutoff and constraints on bonds to hydrogen. NPT stages use a Monte Carlo barostat; NVT stages omit it. The integrator is the Langevin middle integrator with 1 ps^{-1} friction. The 100-ps reporting interval gives 1000 expected frames for a complete 100 ns production trajectory.

MD-derived Q-value extraction

Trajectory input requirements. The processed MD table was generated from the archived 400 K production trajectories. A system was eligible only if it had a production trajectory, a prepared reference structure, and an Amber parameter file. The amino-acid sequence was read from the prepared structure using the first $C\alpha$ atom of each residue, and sequences shorter than 50 residues were excluded.

Native-contact definition. Q-values are computed with MDTraj (47). The topology is loaded from the Amber parameter file, and the native reference is frame 0 of production. The calculation uses protein backbone heavy atoms. A candidate native contact must span more than three residues, have a reference-frame distance below 0.45 nm, and include at least one atom from a hydrophilic residue (Asp, Glu, Gln, Asn, Arg, Lys, or His, including Amber protonation variants). For a native-contact pair (i, j) with reference distance r_{ij}^0 and instantaneous distance r_{ij} , the

per-frame contribution is

$$q_{ij} = \frac{1}{1 + \exp\{\beta(r_{ij} - \lambda r_{ij}^0)\}},$$

with $\beta = 50 \text{ nm}^{-1}$ and $\lambda = 1.8$, following the smooth native-contact form of Best-Hummer-style Q-value analyses (34). The frame-level Q-value is the mean of q_{ij} over all selected native contacts.

Trajectory-level Q-value label. The trajectory-level label is the mean Q-value over the terminal portion of production. For trajectories with more than 300 frames, the implementation uses the larger of 300 frames and one third of the available frames; for shorter trajectories it uses all frames. For a complete 100 ns trajectory with 100-ps output, this is 333 terminal frames, roughly the final 33 ns. In the processed 400 K table, the number of frames used ranges from 66 to 333 (median 333), the number of native contacts from 28 to 728 (median 173), and the sequence length from 58 to 461 residues (median 122). Raw Q-values range from 0.0437 to 0.9994 and are scaled to [0,1] for computational-task training.

Sequence tokenization and encoder inputs

Sequences are tokenized with the ESM2 tokenizer for the selected encoder (14), using truncation and fixed-length padding to 160 residues. In the production protocol, the trainer receives all target and sampled computational-label rows for training, but its evaluation data are filtered to the experimental Tm validation rows.

Hot-encoder input path. The hot-encoder setting keeps tokenized sequences in the training data and updates the ESM2 weights. The frozen-encoder setting runs ESM2 once per target and computational-label sequence to precompute first-token embeddings, after which the optimizer updates only the regression trunk and task heads.

Neural-network architecture

The default encoder is the 8M-parameter ESM2 model, and the model uses its first-token representation as the sequence-level embedding. The shared trunk has dimensions hidden-size $\rightarrow 256 \rightarrow 128 \rightarrow 32$, with ReLU activations and dropout after the first two hidden layers. The Tm head maps the 32-dimensional representation to one scaled scalar prediction.

Target and computational heads. The Tm-only baseline is the encoder, the shared trunk, and the Tm head. Computational-label transfer models keep this target path and add computational heads. The main mutation-effect model uses separate computational heads for the two template-specific tasks. The code also implements computational-head controls, including a single shared mutation-effect head, a template-conditioned head, and a template-specific affine calibration on top of a shared head, to test whether the two template-specific computational labels need separate output calibration.

Architecture sensitivity analyses. Architecture controls include residual, latent, and mixture-style computational-fusion variants, in which the computational path produces a correction to the Tm prediction, a latent interaction between Tm and computational features, or a gate between a target-only expert and a computational-informed expert. The main reported FEP result uses the original shared-trunk architecture with separate template-specific computational heads; the other variants are kept as sensitivity analyses, not as the main claim.

Training objective, optimizer, and checkpoint selection

Each training example is a sequence, a scaled label, and a task identifier, and contributes loss only through the regression head for its task. The per-task loss is Huber loss with $\delta = 1.0$ on the scaled label. The default multi-task objective uses learned task-uncertainty weights (44). For task k with loss L_k and learned log-scale s_k , the contribution is

$$\frac{L_k}{2\exp(2s_k)} + s_k,$$

and the total loss sums the contributions from the tasks present in the minibatch. Fixed computational-loss weights are also implemented and were included in the candidate searches for computational classes where loss balance could shift validation performance.

Optimizer and schedule. The optimizer is AdamW. The default head/trunk learning rate is 3×10^{-4} , the default hot-encoder learning rate is 1×10^{-4} , weight decay is 0.04, dropout is 0.15, batch size is 16, warmup is 100 steps, and the schedule is cosine decay over at most 400 epochs. Checkpoints are evaluated and saved once per epoch, and mixed precision is used when CUDA is available. In the hot-encoder setting, the trainer creates separate optimizer parameter groups for ESM2 parameters and for the freshly initialized trunk/head parameters; in the frozen-encoder setting, ESM2 parameters are non-trainable and no encoder group is used.

Checkpoint selection. Within each run, the best checkpoint has the lowest mean squared error on the experimental Tm validation rows, and early stopping monitors the same metric with patience 15 and threshold 0. Computational-task validation rows are not passed to the trainer in the production protocol. This detail is central to the comparison.

Computational labels shape the learned representation through the training loss, but they do not decide which checkpoint is used for target-task prediction.

Candidate-setting selection and final evaluation

Candidate settings are selected on the experimental Tm validation split, tuning every source independently and separately for the frozen and hot encoder regimes with a two-stage search. Stage 1 fixes learning rate, dropout, and weight decay at defaults and searches the model architecture (shared trunk, residual fusion, or latent fusion) crossed with the auxiliary-head coupling (separate template-specific heads or a template-conditioned shared head). Stage 2 holds the best Stage 1 skeleton and searches, for the hot regime, the encoder learning rate (1×10^{-4} , 5×10^{-5} , 3×10^{-5} , 1×10^{-5}) and, for the frozen regime, the head learning rate (1×10^{-3} , 5×10^{-4} , 2×10^{-4}), each crossed with dropout (0.05, 0.15, 0.30) and weight decay (0.02, 0.10). This staged search covered 30 Stage 1 and 126 Stage 2 configurations for the core sources, and an additional 48 Stage 1 and 168 Stage 2 configurations for the structure-based comparators.

Seeded candidate evaluation. Each candidate is evaluated as a three-seed ensemble on the Tm validation split at $n = 320$ labels per scaffold table, and the single best configuration for each source and regime is then evaluated on the held-out test split as a five-seed ensemble. The run seed controls Python, NumPy, PyTorch, and CUDA randomness when CUDA is available, as well as target-training subsampling in the target-label scaling experiment, computational-row sampling, and computational-label train/validation splitting. The label-count curves use the selected per-source configuration while the number of computational labels is varied over 20, 80, 160, and 320.

Statistics and stored outputs

The primary metric is MAE in degrees Celsius on the 396 held-out test examples. For each condition, the five seeded models produce scaled Tm predictions for every test sequence; these are averaged in scaled-label space and inverse-transformed to degrees Celsius. Root mean squared error, coefficient of determination, and Spearman and Pearson correlations are also computed, but the figures emphasize MAE because it reads directly as an average temperature error.

Bootstrap intervals. Single-condition confidence intervals come from nonparametric bootstrap resampling over test examples. The absolute-error vector from the five-seed ensemble is resampled with replacement 10,000 times using a fixed bootstrap seed, and the reported 90% interval is the 5th to 95th percentile range of the bootstrapped MAE values. The 90% interval is used for consistency with the directional paired comparisons in the main text; the stored bootstrap summaries allow other interval levels to be recomputed. Paired comparisons use the same bootstrap indices for the candidate and reference absolute-error vectors. The reported ΔMAE is candidate minus reference, so negative values mean lower error than the reference.

Stored run summaries. Each training and evaluation call writes a structured summary recording the command-line arguments, relevant environment variables, resolved model settings, per-label-count metrics, bootstrap intervals, per-example absolute errors, and paired bootstrap summaries. Compact tables derived from these summaries feed the main computational-label comparison, the frozen-encoder controls, the computational-head controls, and the computational-combination controls.

Interpreting the label-count curves

Target-label scaling reference. The target-label scaling curve is a positive control on the pipeline. It tests whether the same model family and evaluation code recover the expected gain when more true experimental Tm labels are available. The FEP computational-label curve tests something different. It asks whether more labels from a physically related but distinct computed quantity improve the same held-out Tm endpoint, while the FEP-matched MD native-contact curve tests whether a structural stability observable responds to a matched simulation-acquisition design. The decisive comparisons are therefore not “simulation versus no simulation,” but whether the simulations cover relevant sequence changes and whether the physical quantity improves models selected with Tm validation data.

Supplementary figure data tables

The supplementary figures are generated from tabular computational-label files derived from the same run summaries as the main figures. The processed computational tables, rendered supplementary figures, and figure-generation code will be distributed with the public analysis archive.

Fig.-to-data and code mapping

The manuscript figures are regenerated from compact result summaries and computational-label tables rather than from scratch run directories. The mapping below is the first place to check when auditing or extending a figure.

Table 2. Manuscript Fig.-to-data mapping. The table identifies the processed summaries and scripts used to regenerate each display item.

Display item	Inputs	Script	Primary comparison
Fig. 1	Conceptual protocol	plot/make_outline_figures.py	Multi-task training with Tm-based model selection
Fig. 2	Simulation-plan schematic; design-by-regime MD effects; frozen label-count sweeps	plot/make_outline_figures.py	Simulation-design axis: matched local scans strengthen frozen native-contact transfer
Fig. 3	Source-by-regime map; focused FEP/MD contrasts; hot label-count sweeps	plot/make_outline_figures.py	Physical-observable axis: FEP transfers in both regimes, MD native-contact only frozen
Supplementary Figs. 1 to 5	Supplementary manifest and generated TSV tables	plot/make_supplementary_figures.py	Data-set composition, candidate-setting searches, model controls, count-sweep controls, and MD-feature controls

Generated supplementary tables. The final computational-label screen settings used in Figs. 3 and 4 and the frozen-encoder controls are recorded in generated supplementary tables, which list the selected architecture, validation-selected setting, computational-label count, validation MAE, held-out test MAE, bootstrap interval, and run-summary reference for each condition. A panel-level mapping table for all supplementary figures is included in the public analysis archive.

Supplementary Figs.

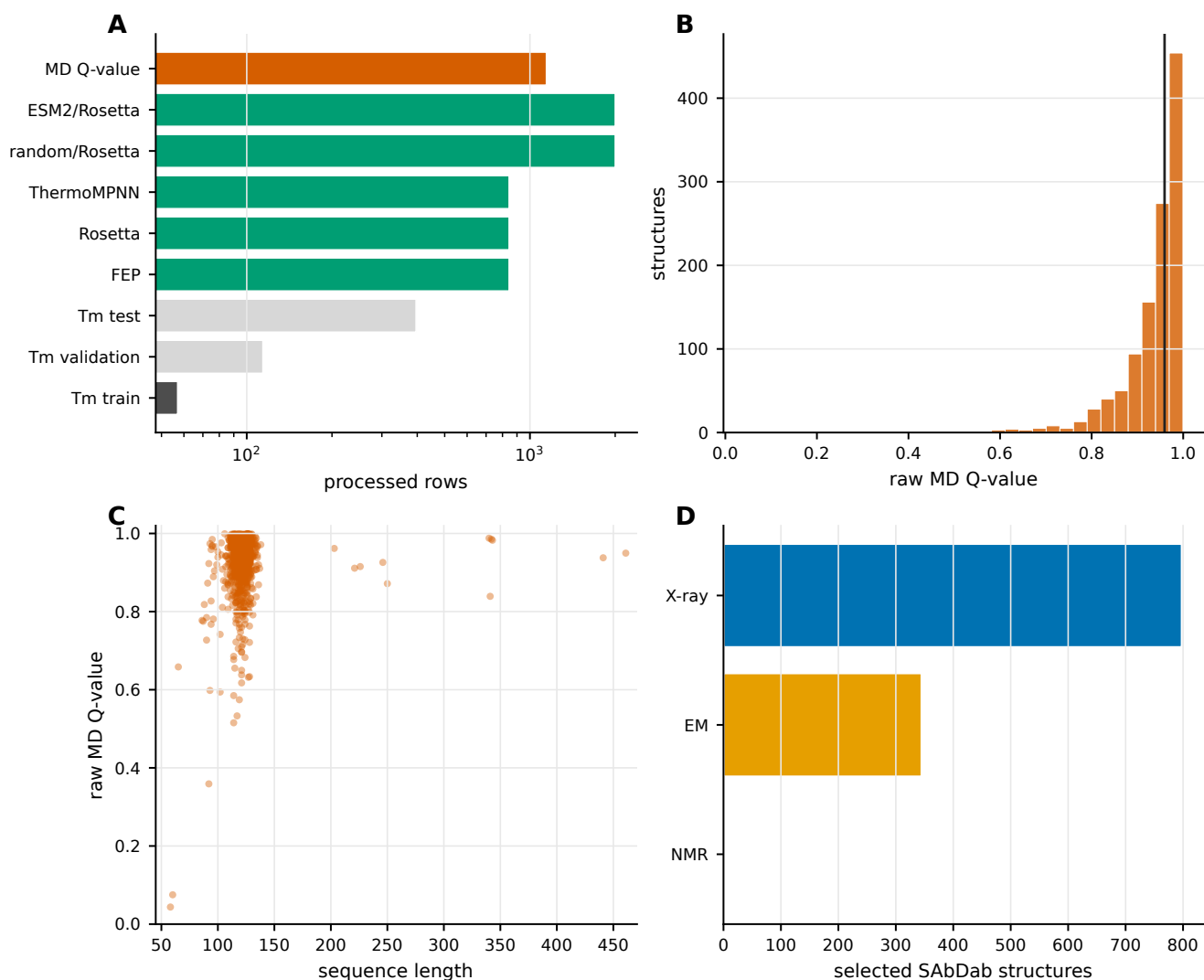


Fig. 4. Supplementary Fig. 1. Processed data sets and MD-derived Q-value labels. (a) Processed target and computational-label row counts used in the reported experiments. (b) Distribution of raw MD-derived Q-values before [0,1] scaling. (c) Relationship between sequence length and raw MD-derived Q-value. (d) Experimental methods for the selected SAbDab structures used in the 400 K MD panel.

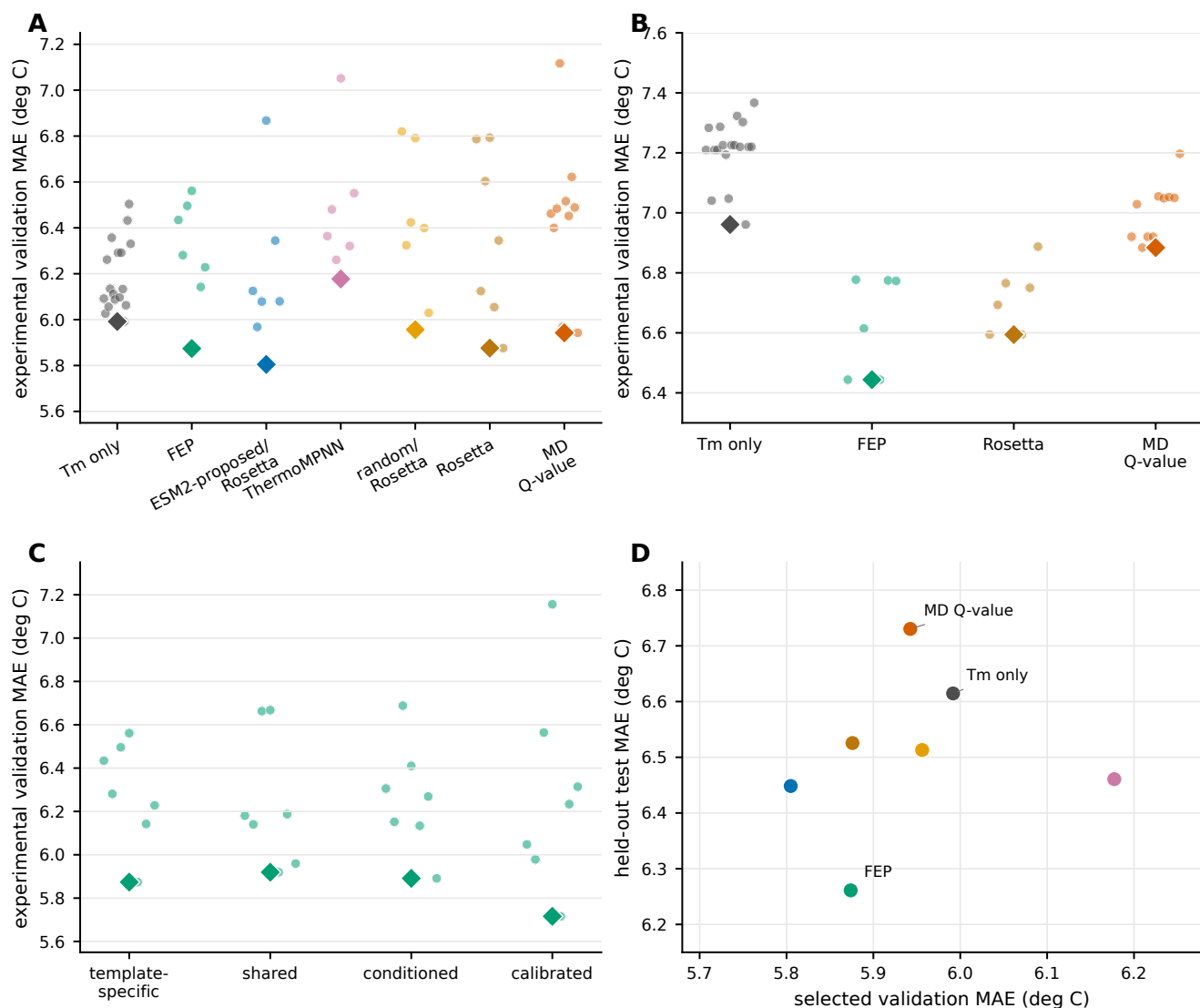


Fig. 5. Supplementary Fig. 2. Candidate-setting searches and target-validation selection. (a) Experimental validation MAE values for hot-encoder computational-label comparisons. (b) Experimental validation MAE values for frozen-encoder controls. (c) Candidate computational-head designs for FEP mutation free-energy labels. (d) Relationship between selected validation MAE and held-out test MAE for the final computational-label comparison. Diamonds mark the best validation setting in panels a to c.

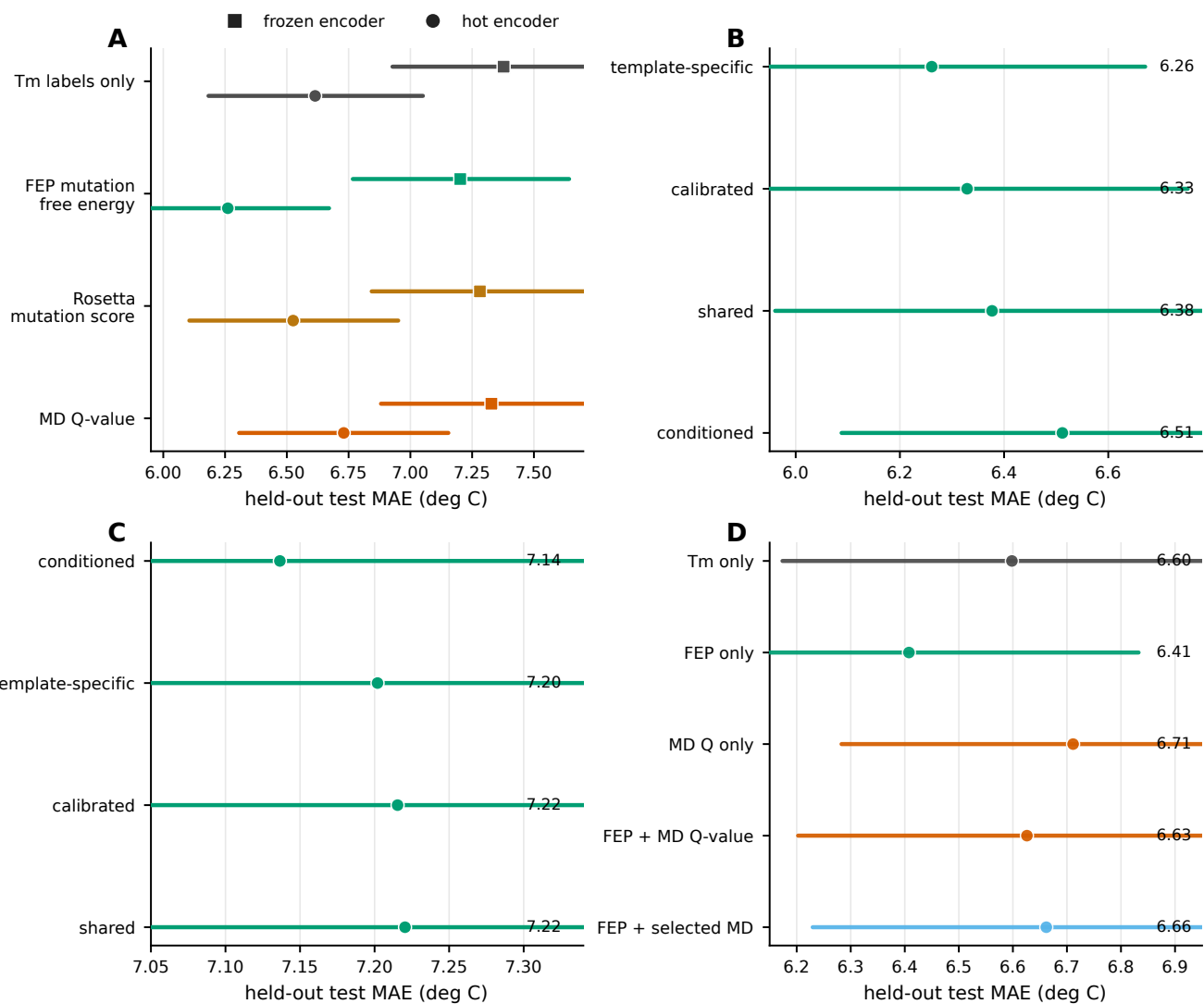


Fig. 6. Supplementary Fig. 3. Model and computational-combination controls. (a) Frozen-encoder and hot-encoder comparisons for selected computational labels. (b) Hot-encoder FEP computational-head controls. (c) Frozen-encoder FEP computational-head controls. (d) Computational-combination controls comparing Tm-only, FEP-only, MD Q-value-only, and combined-computational-label settings.

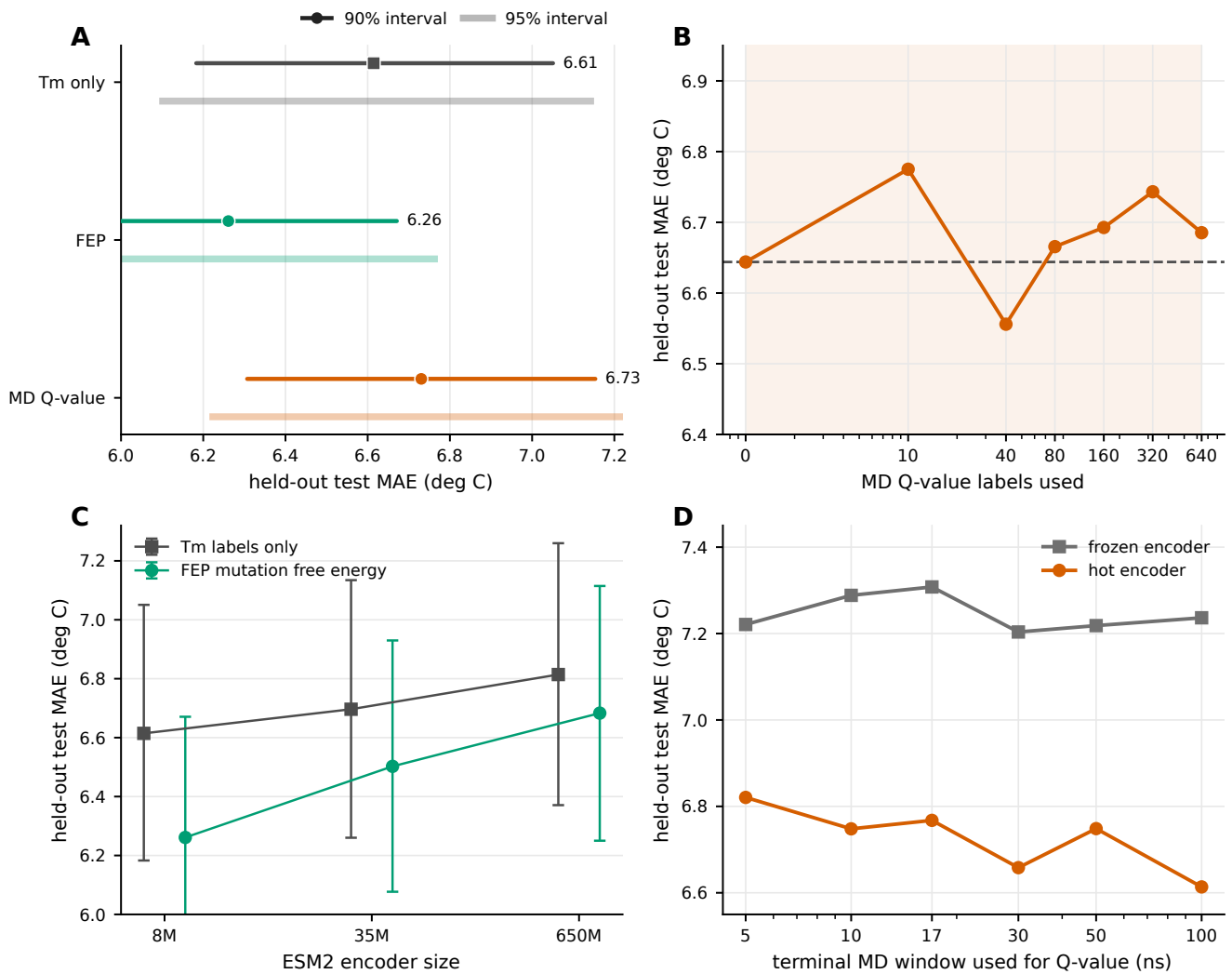


Fig. 7. Supplementary Fig. 4. Interval, scaling, encoder-size, and MD-window controls. (a) Final Tm-only, FEP, and MD Q-value conditions shown with 90% and 95% bootstrap intervals. (b) MD Q-value label-count curve after selecting the candidate setting separately for each label count on the experimental validation split. (c) ESM2 encoder-size controls for Tm-only and FEP settings. (d) MD trajectory-window controls for Q-value labels.

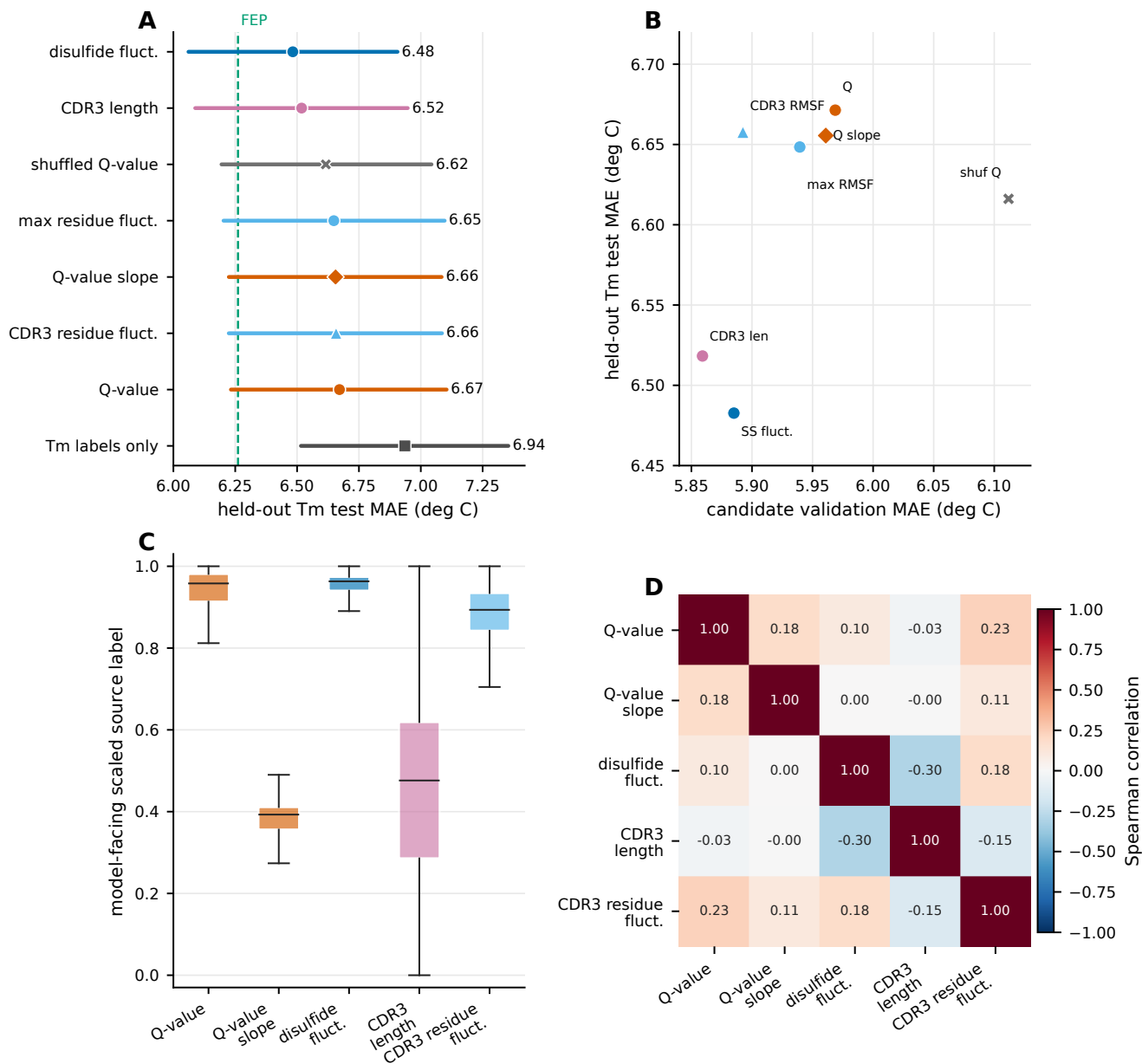


Fig. 8. Supplementary Fig. 5. Descriptor controls for MD-derived computational labels. (a) Held-out test MAE values for selected MD-derived and nanobody-specific descriptor controls. The dashed line marks the FEP reference. (b) Relationship between candidate validation MAE and held-out test MAE for descriptor controls that were carried forward to final evaluation. (c) Distributions of the model-facing scaled computational labels for the selected descriptors. (d) Spearman correlations among the selected computational labels over shared sequences.